Lab 10 –

Introduction to Files

Programming Exercise:

1.Write a function stats() that takes one input argument: the name of a text file. The ;function should print, on the screen, the number of lines, words, and characters in the file .your function should open the file only once stats('example.txt') >>> line count: 3 word count: 20 character count: 98

Source Code:

```
# question 1
def stats(filename):
    try:
        with open(filename, 'r') as file:
            content = file.read()
        line_count = content.count('\n') + 1
        word_count = len(content.split())
        char_count = len(content)
        print(f"Line count: {line_count}")
        print(f"Word count: {word_count}")
        print(f"Character count: {char_count}")
    except FileNotFoundError:
        print(f"The file {filename} was not found.")
stats(r"C:\New folder\abc.txt")
```

Output:

```
C:\Users\laiba\PycharmProjects\pythonProje
Line count: 2
Word count: 14
Character count: 72

Process finished with exit code 0
```

2. Implement function distribution() that takes as input the name of a file (as a string). This one-line file will contain letter grades separated by blanks.Your function should .print the distribution of grades, as shown

distribution('grades.txt') >>> students got A 6-students got A 2 +students got B 3 students got B 2-students got B 2 students got C 4-student got C 1 students got F 2.

Source Code:

```python
# question 2
from fileinput import filename
def distribution(filename):
    try:
        with open(filename,'r') as file:
            grades = file.read().split()
        grade_count = {'A':0,'B':0,'C':0,'F':0}
        for grade in grades:
            if grade in grade_count:
                grade_count[grade] += 1

        cumulative_count = 0
        for grade in ['A','B','C','D']:
            count = grade_count[grade]
            cumulative_count += count
            print(f'students got {grade} {count}')
            if cumulative_count > count:
                print(f'-students got {grade} {cumulative_count - count}')
            if cumulative_count < sum(grade_count.values()):
                print(f'+students got {grade} {cumulative_count}')
    except FileNotFoundError:
        print(f"the file does not exist {filename}")
    except Exception as e:
        print("an error occured",e)
distribution(r'C:\New folder\grades.txt')
```

Output:

```
C:\Users\laiba\PycharmProjects\pythonProject\.v
students got A 3
+students got A 3
students got B 3
-students got B 3
+students got B 6
students got C 2
-students got C 6
+students got C 8
an error occured 'D'

Process finished with exit code 0
```

3. Implement function duplicate() that takes as input the name (a string) of a file in the .current directory and returns True if the file contains duplicate words and False otherwise duplicate('Duplicates.txt') >>> True duplicate('noDuplicates.txt')

>>> False The function abc() takes the name of a file (a string) as input. The function should .

Source Code:

```python
# question 3
def duplicate(filename):
    try:
        with open(filename,'r') as file:
            content = file.read()
            words = content.split()
            seen_words = set()
            for word in words:
                if word in seen_words:
                    return True
                seen_words.add(word)
        return False
    except FileNotFoundError:
        print('the file does not exist.')
        return False
result = duplicate(r'C:\New folder\grades.txt')
print("contains duplicate:",result)
```

Output:

```
C:\Users\laiba\PycharmProjects\pythonProject\
contains duplicate: True

Process finished with exit code 0
```

4. open the file, read it, and then write it into file abc.txt with this modification: Every .'occurrence of a four-letter word in the file should be replaced with string

'xxxx abc('example.txt') >>> Note that this function produces no output, but it does create file abc.txt in the current folder.

Source Code:

```python
# question 4
def abc(filename):
    try:
        with open(filename,'r') as file:
            content = file.read()
        words = content.split()
        modification = ['xxxx' if len(words) == 4 else word for word in words]
        modified_content = ' '.join(modification)
        with open('abc.txt','w') as output_file:
            output_file.write(modified_content)
        print("file abc.txt has been created with changes")
    except FileNotFoundError:
        print("the file does not exist")

abc(r'C:\New Folder\abc.txt')
```

Output:

```
C:\Users\laiba\PycharmProjects\pythonProject\.venv\Scripts\python.exe C:\Use
file abc.txt has been created with changes

Process finished with exit code 0
```